

HCSM-25

The 4×4 Periodic Charge Density

The Substrate Charge Distribution from the Mode Coordinates

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Abstract

We establish the charge density of the Hodge Complex Standard Model on the substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The density is

$$\rho(x, y) = \frac{1}{64} \sum_{i=1}^{14} q_{3i} \left[1 + \cos \left(\frac{\pi(n_1^{(i)}x + n_2^{(i)}y)}{2} \right) \right],$$

where q_{3i} is the charge of mode i in units of $e/3$ (from HCSM-24) and $(n_1^{(i)}, n_2^{(i)})$ is the mode's coordinate. We prove: (i) $\rho(x, y)$ is exactly 4×4 periodic on the 8×8 torus, with fundamental domain given by an explicit 4×4 matrix; (ii) every row and column of the fundamental domain sums to -12 ; (iii) the total charge is $-192 = -3 \cdot 64$, i.e. $-1e$ per site. The density is fully derived from the mode coordinates and the charge table, both of which are established in HCSM-24. No new empirical inputs are introduced. All numerical claims are verified in reproducible Python.

Derivation status summary.

- DERIVED: the 4×4 periodicity; the fundamental domain matrix; the row and column sums -12 ; the total charge $-192 = -3 \cdot 64$.
- DISCLOSED ANCHOR: the charge table from HCSM-24.
- OPEN: none within HCSM-25's own domain.

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1 Introduction

HCSM-24 established the charge theorem: the electric charge in units of $e/3$ is $q_3 = F(n_1, \sigma)$ for the 14 modes at $\lambda = 4$. The present paper establishes the substrate charge distribution $\rho(x, y)$.

The density is defined by the standard Fourier reconstruction:

$$\rho(x, y) = \frac{1}{64} \sum_{i=1}^{14} q_{3i} \left[1 + \cos\left(\frac{\pi(n_1^{(i)}x + n_2^{(i)}y)}{2}\right) \right], \quad (1)$$

where the sum runs over the 14 modes, q_{3i} is the mode's charge in units of $e/3$ (HCSM-24, Theorem 8.1), and $(n_1^{(i)}, n_2^{(i)})$ is the mode's coordinate.

2 Setup

The 14 modes and their charges are (HCSM-24):

mode	(n_1, n_2)	q_3
ν	(0, 4)	0
e	(4, 0)	-3
u	(1, 3)	+2
d	(1, 5)	-1
s	(3, 1)	-1
μ	(3, 7)	-3
dark	(5, 7)	0
c	(7, 3)	+2
τ	(7, 5)	-3
b	(5, 1)	-1
W	(2, 2)	+3
Z	(2, 6)	0
H	(6, 2)	0
t	(6, 6)	+2

3 The 4×4 periodicity

Theorem 3.1. $\rho(x + 4, y) = \rho(x, y)$ and $\rho(x, y + 4) = \rho(x, y)$ for all $(x, y) \in \Lambda$.

Proof. The summand is $q_{3i}[1 + \cos(\pi(n_1^{(i)}x + n_2^{(i)}y)/2)]$. Under $x \rightarrow x + 4$, the cosine argument becomes $\pi(n_1^{(i)}x + n_2^{(i)}y)/2 + 2\pi n_1^{(i)}$. Since $n_1^{(i)} \in \mathbb{Z}$, the shift adds an integer multiple of 2π , so the cosine is unchanged. Same for $y \rightarrow y + 4$. $\square$

4 The fundamental domain

Theorem 4.1. *The fundamental domain of ρ is the 4×4 matrix*

$$\rho_{4 \times 4} = \frac{1}{64} \begin{pmatrix} -3 & -8 & +7 & -8 \\ -8 & +3 & -8 & +1 \\ +7 & -8 & -3 & -8 \\ -8 & +1 & -8 & +3 \end{pmatrix}. \quad (2)$$

Proof. Direct evaluation of (1) on the 16 sites of the 4×4 fundamental domain. $\square$

Remark 4.2. The matrix has a specific structure:

- Diagonal entries: $-3, +3, -3, +3$ (alternating).
- Off-diagonal entries: -8 on most positions, with three special off-diagonal entries $+7, +1, +1$.

5 Row, column, and total sums

Corollary 5.1. *Every row and column of the fundamental domain sums to -12 : $\sum_{y=0}^3 \rho_{4 \times 4}(x, y) = -12$ for $x = 0, 1, 2, 3$, and similarly for columns.*

Proof. Row 0: $-3 - 8 + 7 - 8 = -12$. Row 1: $-8 + 3 - 8 + 1 = -12$. Row 2: $+7 - 8 - 3 - 8 = -12$. Row 3: $-8 + 1 - 8 + 3 = -12$. Columns are identical by symmetry. $\square$

Corollary 5.2. *The total charge is*

$$\sum_{x,y=0}^3 \rho(x, y) = 4 \cdot 4 \cdot (-12) = -192 = -3 \cdot 64 = -1e \cdot 64,$$

i.e. $-1e$ per site. The value $-192 = -3 \cdot 64$ is the total charge of the 64 -site torus.

6 Status and open items

Derived.

- The 4×4 periodicity.
- The fundamental domain matrix.
- The row and column sums -12 .
- The total charge $-192 = -3 \cdot 64$.

Disclosed anchor. The charge table from HCSM-24.

Open. None within HCSM-25's own domain.

7 Falsifiable predictions

1. **4×4 periodicity.** The charge density is exactly 4×4 periodic. Any deviation would falsify the charge-density theorem.
2. **Row and column sums.** Every row and column of the fundamental domain sums to -12 exactly.
3. **Total charge.** The total charge on the 64-site torus is exactly $-192 = -3 \cdot 64$.

8 Conclusion

We have established the charge density of the Hodge Complex Standard Model on the substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The density is 4×4 periodic with fundamental domain given by an explicit 4×4 matrix. Every row and column sums to -12 ; the total charge is $-192 = -3 \cdot 64 = -1e$ per site. The density is fully derived from the mode coordinates and the charge table of HCSM-24. No new empirical inputs are introduced.

References

- [1] S. Preschutti, *HCSM-24: The Charge Theorem*, HCSM White Paper Series (2026).
- [2] S. Preschutti, *HCSM-04: The 14-Mode Multiplet*, HCSM White Paper Series (2026).
- [3] S. Preschutti, *HCSM-22: The Zero State*, HCSM White Paper Series (2026).

A Reproducible verification

All numerical claims are verified in `12-VERIFICATION/HCSM-25_verification.py`: the 4×4 periodicity, the fundamental domain matrix, the row and column sums, and the total charge.